CONTROL PHILOSOPHY

FOR

PRODIGY CHILLERS

TABLE OF CONTENTS

1. Introduction

The controller of typical chiller plant does various functions from precisely controlling chilled water temperature by loading/unloading compressor and managing EXV to logging data and alarm history. Off course while doing so it has to ensure safety by monitoring and proactively controlling some parameters. KCPL K smart controller can even manage allied plant equipment like CT fan, Pumps and valves. Satisfying all these requirements made feasible by today's fast microprocessor based controllers wherein many parallel control algorithms execute at the same time. Thus it's not straight forward affair to completely explain control philosophy of a chiller controller. It can be simplified by breaking chiller operations into its various states as well as looking at various parallel control loops one at a time.

2. List of Major Components in Chiller

Sr. No	Major Components
1	Screw Compressor
2	Evaporator
3	Condenser coils
4	Electronic Expansion Valve
5	Oil Cooler & Economizer (Optional)

3. Control Philosophy

The PRODIGY screw chiller operation can be divided as below sub-parts:

- ✓ Start up and Shut down sequence.
- ✓ Loading/Unloading of compressor to match external load.
- ✓ Proactive controlling and safety trips.
- ✓ Oil cooler operation.(Optional)
- ✓ Data and alarm history logging.
- ✓ Refrigerant flow control by electronic expansion valve.

Most of above operations happens simultaneously, with controller simultaneously executing all control loops. These are discussed here in detail one by one.

4.1. Start-up and Shut down: Working & Control Philosophy:-

The compressor is driven by semi-hermetic induction motor, which can be offered with various starter options such as Star-delta, Soft Starter, VFD etc. These starter panels and control panel is mounted on chiller. For details refer Control wiring diagram of control panel, starter panel and Interfacing wiring diagram.

Chiller can be operated in local mode & remote mode local/remote mode selection shall be done from chiller controller. Remote mode operation shall be initiated from BMS/Plant-Manager (PM) by start/stop command to chiller control panel. During manual mode BMS/Plant-Manager (PM) intervention shall be bypassed.

Refer algorithm for startup and shut down for detailed sequence.

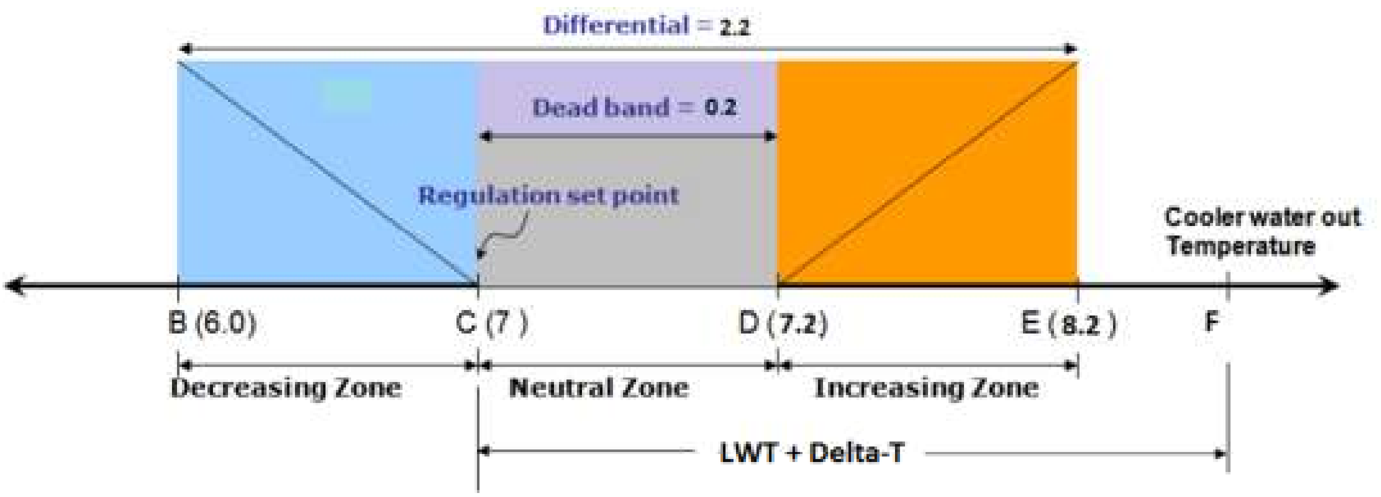

Fig 1. Shows the Increasing zone/Neutral zone/Decreasing zone calculated according to set points.

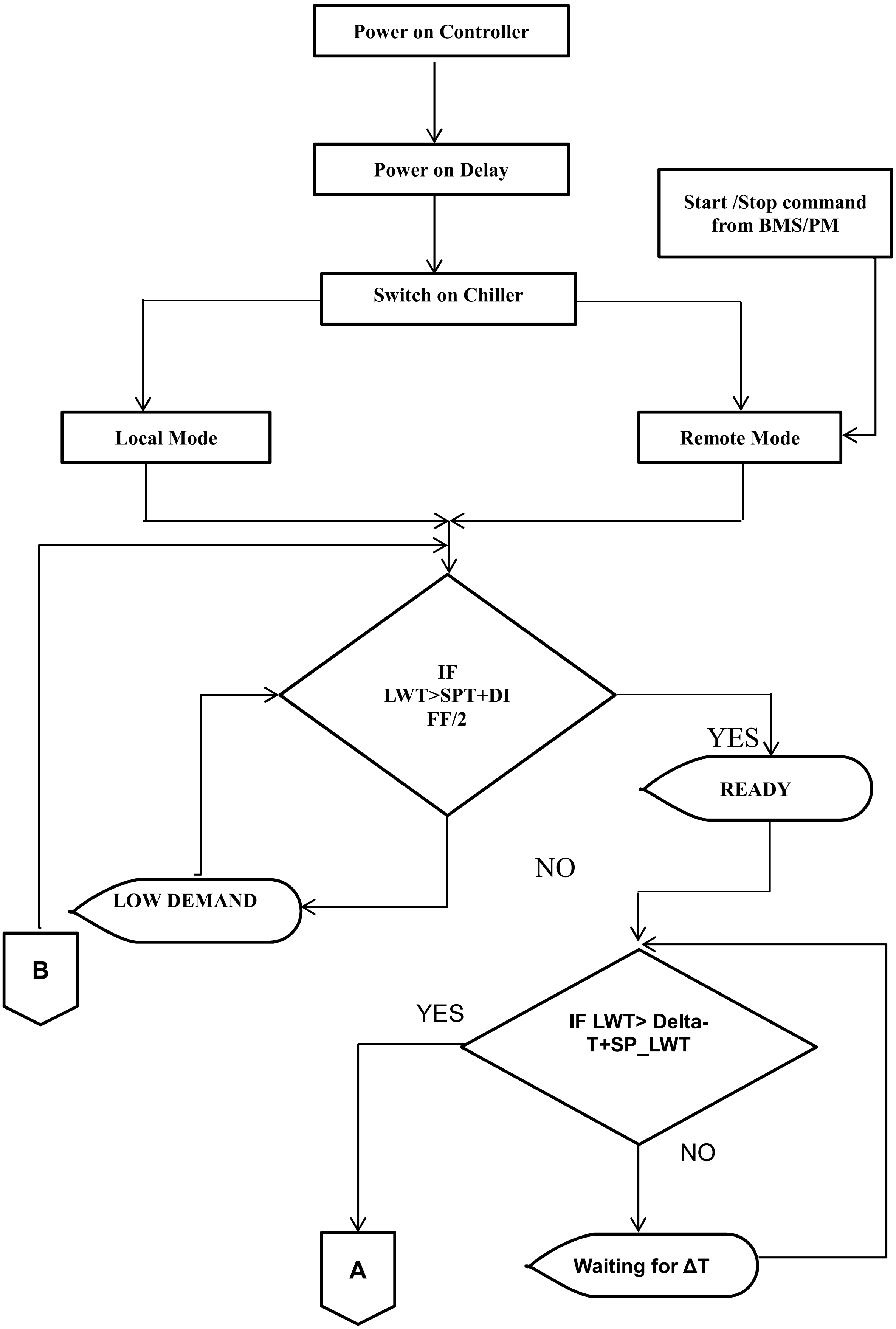

Fig 3 Algorithm for Startup and Shut down for Detailed Sequence (1)

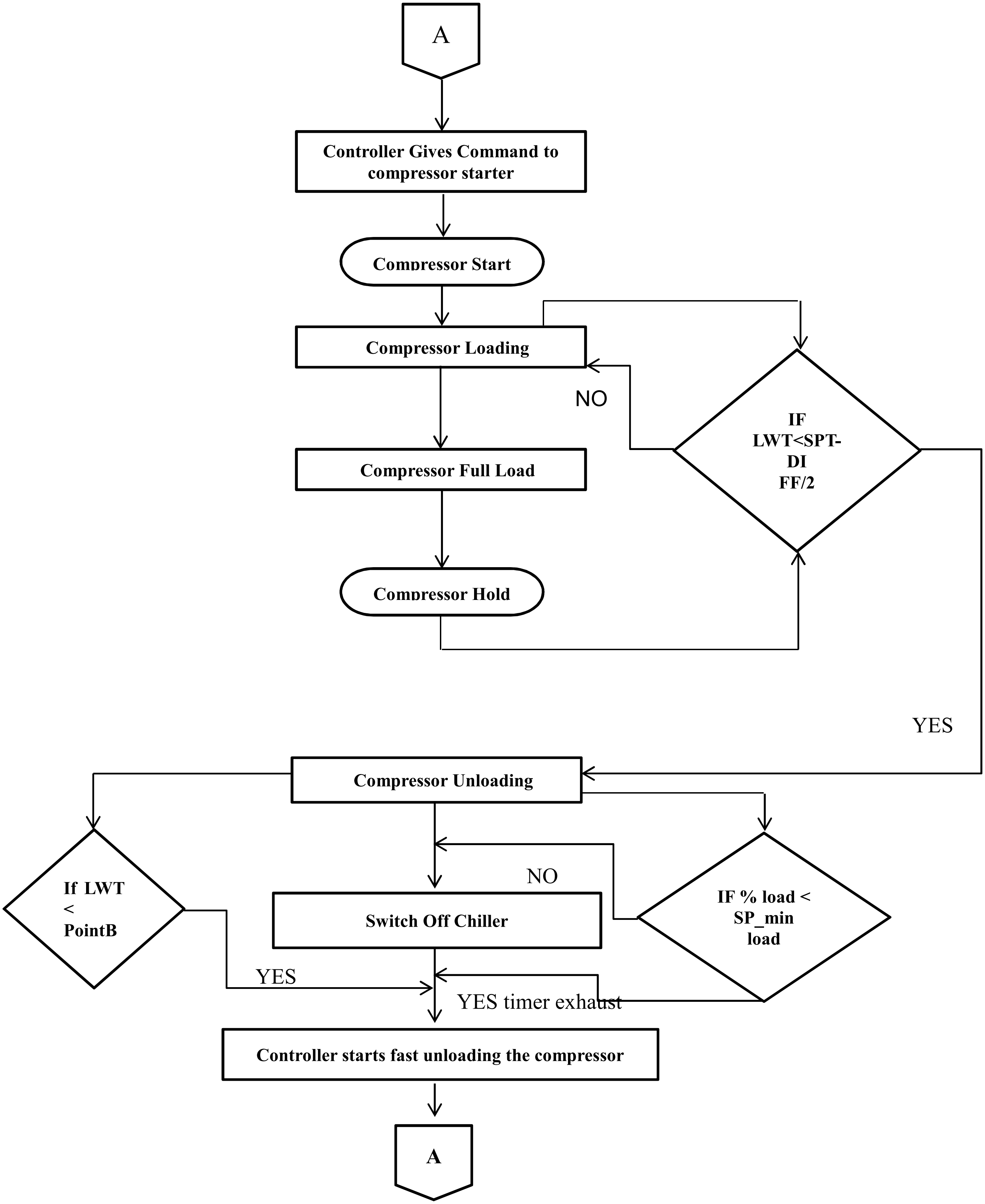
A
Controller Gives Command to compressor starter
Compressor Start
Compressor Loading
NO
IF LWT<SPT-DIFF/2
Compressor Full Load
Compressor Hold
YES
Compressor Unloading
If LWT < PointB
NO
Switch Off Chiller
IF % load < SP_min load
YES
YES timer exhaust
Controller starts fast unloading the compressor
A

Fig 4 Algorithm for Startup and Shut down for Detailed Sequence (2)

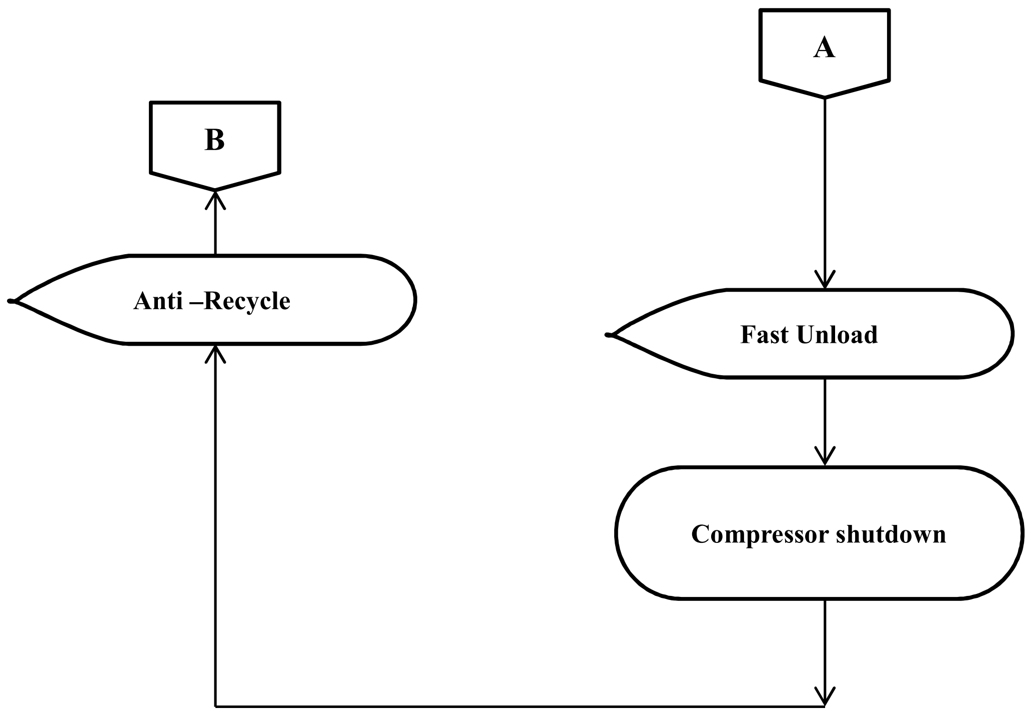

Fig 5 Algorithm for Startup and Shut down for Detailed Sequence (3)

4.2. Loading/Unloading Control: Working & Control Philosophy:-

Cooling capacity control is achieved by means of a slide valve mechanism controlled by microprocessor system through two solenoid coils and by varying speed by VFD. Each unit has infinitely variable (step less) capacity control from 100% down to 35%. This modulation allows the compressor capacity to match the building-cooling load. The result is a decrease in chiller energy costs, particularly at the part-load conditions at which the chiller operates most of the time.

KAS chillers are equipped with Electronic expansion valve, which quickly reacts to load variations and ensures precise, metered refrigerant flow into the evaporator.

Following diagram illustrates different zones viz; Increasing zone, Decreasing Zone and Dead band created by three set points – Target set pt., dead band & differential.

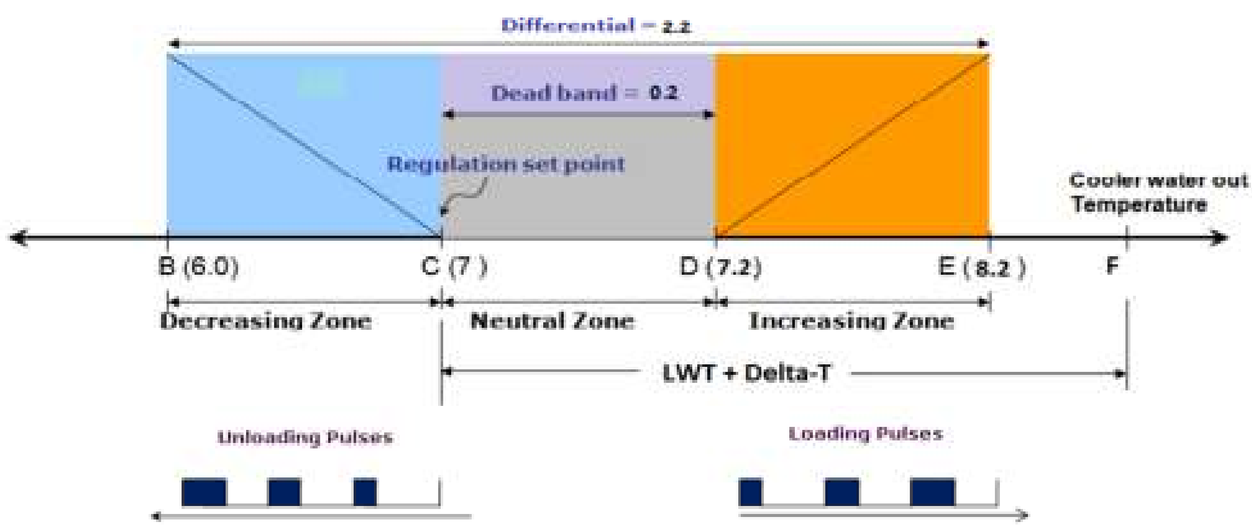

Fig.6 Loading Unloading PID Logic

The above diagram is self-explanatory, if the chilled water temperature is in increasing controller gives the loading pulses proportional to its deviation from neutral zone. If it's above the differential band (above pt. E) then it gives maximum size pulses. Similarly if the chilled water temperature is in decreasing zone controller gives the unloading pulses proportional to its deviation from neutral zone. If chilled water temperature drops below differential band (below pt. B), controller shuts the compressor off known as "Routine Shutdown" and restarts only when temperature rises above point F (LWT SP+ Delta-T Set point).

4.3. Proactive Controlling and Safety Trips:-

Various digital interlocks or safeties are provided for chiller, some of them are Auto reset, Means if the unsafe condition corrected automatically trip on controller vanishes while some of them are Manual reset, means operator has to reset the alarm then only chiller can be started. Following is list of digital safeties provided.

Sr. No	Interlock	Reset	Delay	Active During
1	DI CHILLER WATER FLOW	Auto	5 SEC	After the switch ON
2	DI VOLTAGE PROTECTION DEVICE	Manual	1 SEC	Throughout running of chiller
4	DI TRANSITION OK	Manual	10 SEC	Chiller starting
5	DI STARTER FAULT	Manual	1 SEC	Throughout running of chiller
6	DI COMPRESSOR WINDING	Manual	1 SEC	Throughout running of chiller

Apart from above digital safeties, various analog safeties are also provided. For some analog safeties instead of directly tripping compressor, K- smart controller utilizes the pro-active controlling. That means if any of the parameters goes to unsafe condition (Yellow zone) from healthy state (Blue zone); it first unloads the compressor and tries to correct the situation. If parameter returns again to healthy zone, it again loads it normally. If at all parameter goes to critical or risky zone (Red zone) it immediately stops the compressor.

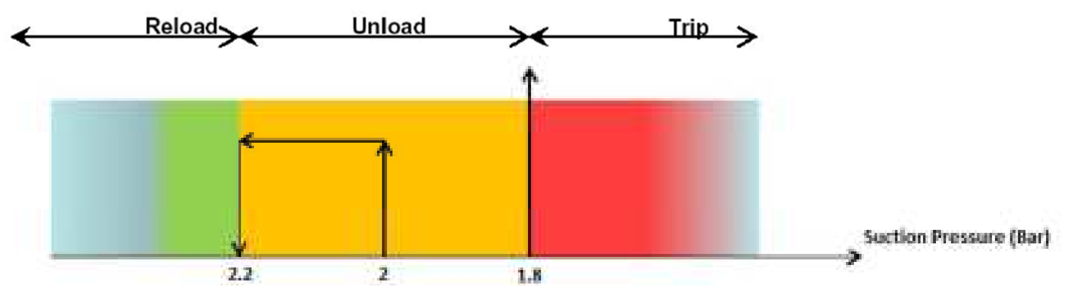

Fig.7 Proactive Controlling

4.3.1. List of Analog Safeties:-

Sr. No	Safety	Description	Proactive Controlling Applicable	Delay	Active when
1	PROBE ALARM	GENERATES WHEN PROBE BECOMES FAULTY	NO	10 SEC	After the power up delay
2	CURRENT ALARM-				
	a).LOW CURRENT TRIP	GENERATES WHEN CURRENT VALUE IS BELOW SET LIMITS	NO	2 SEC	During Compressor Running
	b).HIGH CURRENT TRIP	GENERATES WHEN CURRENT VALUE IS ABOVE SET LIMITS	YES	0 SEC	During Compressor Running
3	PRESSURE ALARM				
	a). VERY LOW SUCTION PRESSURE	GENERATES WHEN SUCTION PRESSURE IS BELOW MIN SET LIMIT	YES	2 SEC	During Compressor Running
	b). LOW SUCTION PRESSURE	GENERATES WHEN SUCTION PRESSURE IS BELOW LOW SET LIMIT	YES	15 SEC	During Compressor Running
	c). LOW OIL DIFF PRESSURE	GENERATES WHEN OIL PRESS IS BELOW SET LIMIT	NO	30 SEC	During Compressor Running
	d). HIGH DISCHARGE PRESSURE	GENERATES WHEN DISCH PRESS IS ABOVE SET LIMIT	YES	0 SEC	During Compressor Running
4	TEMPERATURE ALARM				
	a). ANTIFREEZE ALARM	GENERATES WHEN LWT IS BELOW ANTIFREEZE SET LIMIT	NO	0 SEC	During Compressor Running
	b). HIGH SUCTION TEMPERATURE	GENERATES WHEN SUCTION TEMP IS ABOVE SET LIMIT	NO	300 SEC	During Compressor Running
5	COMMUNICATION ALARM				
	a). EXP BOARD OFFLINE	GENERATES WHEN EXP BOARD NOT COMUNICATING WITH CONTROLLER	NO	25 SEC	After the power up delay
6	DP SWITCH				
	ACROSS EVAPORATOR	GENERATES WHEN BELOW LOW SET LIMIT	YES		

5.3.2. List of Minor and Major Alarms:-

MINOR	MAJOR
Probe Alarm	Low Current Trip
High Discharge Temperature	High Current Trip
Expansion Board Offline	Pressure Alarm –Very Low Suction Press
	Low Oil Differential Pressure
	High Discharge Pressure
	Anti-Freeze Alarm
	High Suction Temperature
	No Oil Flow

4.4. Oil Cooler & Economizer Operation (Optional):-

To control the oil temperature within certain band separate oil cooler PHE in oil sump & compressor body are provided. Oil Cooler On/Off is governed by following conditions.

- ✓ Oil Cooler On:

Oil temperature < *Oil cooler SP + ON Differential*

- ✓ Oil Cooler Off:

Oil temperature > Oil Cooler SP.

Similarly for Economizer, there is target load set point and ON/OFF differential which controls the opening closing of expansion valve and in turn control the water flow through PHE.

4.5. Data and Alarm History Logging:-

Apart from above parallel control loops, controller saves the 99 alarms and important Parameters at the time of each alarm. These alarms can be viewed on the controller screen one by one.

4.6. Refrigerant Flow Control by Electronic Expansion Valve:-

Electronic expansion valve is used to control the refrigerant flow with in desired limits. At startup EXV is operated on suction pressure to maintain suction pressure at desired value. After the startup sequence proactive control of EXV is done on suction pressure and suction superheat.

- ✓ EXV Open:

Suction Superheat > Superheat set point + Dead Zone +ve band.

- ✓ EXV Close:

Suction Superheat < Superheat set point - Dead Zone -ve band.

EXV is not operated in Dead-zone region. While maintaining Suction superheat controller also looks for suction pressure to be maintained at set value.